

Institutional Profiles Under a Common Prudential Framework:

Evidence from Brazilian Credit Cooperatives and Banks

Award category: Best Young and Emerging Scholar Paper

Arthur Gomes Nery, Thiago de Oliveira Victorino and Rodrigo Lima Rangel

Organization of Brazilian Cooperatives (OCB System), Brazil

Abstract

In Brazil, a credit cooperative that grows past a size threshold computes capital, leverage and risk under the same prudential methodology as a commercial bank, while its cooperative acts remain untaxed. We ask whether the cooperative form still leaves a distinct imprint on the balance sheet. The data are a quarterly panel of the cooperatives and banks under the full methodology from 2017 to 2024: 130 singular credit cooperatives and 144 banks. Eleven balance-sheet and performance measures are related to a cooperative dummy, controlling for size and time, on a matched sample and across five comparison groups. The cooperative form leaves a clear imprint, in three results. First, at comparable size cooperatives operate at lower cost and show steadier returns, and nationally they lend a larger share of their assets than banks, a difference that does not survive matching within region. After tax they retain about two percentage points more of their return, of which the tax exemption explains about two fifths; before tax the typical cooperative still earns more, though the mean gap is not established. Second, capital differs modestly in level and markedly in dispersion. The mean gap comes from small banks with capital ratios above fifty percent and roughly halves against deposit-funded banks; in the lower part of the capital distribution, cooperatives hold more capital than banks at the same quantile and provision no less. Third, two differences often reported, lower provisioning and narrower margins, disappear once the comparison group is restricted to banks. Whether these differences bear on the territorial role of cooperative lenders is the claim the data cannot yet settle. What they do show is that a benchmark calibrated on the average institution may describe the locally lending cooperative poorly.

Keywords: credit cooperatives; prudential regulation; Basel; proportionality; Brazil.

JEL: G21, G28, P13.

1 Background

Prudential regulation has converged on a common template. Basel III supplies the architecture for capital, leverage and risk measurement, and national supervisors adapt it to local conditions [1]. Brazil sorts authorised institutions into five segments for the proportional application of prudential rules [2]; Figure 1 shows how cooperatives and other institutions divide between the two methodologies. Segments 1 to 4 apply the full methodology, while Segment 5 comprises institutions with total exposure below 0.1 percent of GDP that opt for a simplified method of computing minimum capital, governed since February 2018 by Resolution 4.606/2017 [3]. Commercial, multiple, investment and exchange banks are excluded from S5 at any size. Institutions outside S5 apply the *Patrimônio de Referência* capital definition, minimum requirements and risk-weighted-asset computation of Resolutions 4.955 and 4.958, in force since 3 January 2022, and of their 2013 predecessors before that [4, 5]. Loan classification and minimum provisioning follow Resolution 2.682/1999 throughout the window [6].

A credit cooperative that grows past the threshold, or declines the simplified option, therefore computes capital under the same definition, minimum ratios, standardised risk weights and provisioning rules as a commercial bank. A few requirements differ by segment: the liquidity ratios and the systemic buffer apply to S1 only, and the Basel leverage ratio to S1 and S2 [5, 7–9]. Those segments contain banks only, and Section 4.6 shows that excluding them changes little.

One rule difference remains. The results of cooperative acts, the operations between a cooperative and its members, are outside taxable income under Law 5.764/1971 and exempt from the social contribution on profit under Law 10.865/2004, while banks pay both taxes [10, 11]. The common methodology thus equalises the prudential rules and leaves the fiscal ones apart, so we measure profitability before and after tax and treat the difference as a channel of its own.

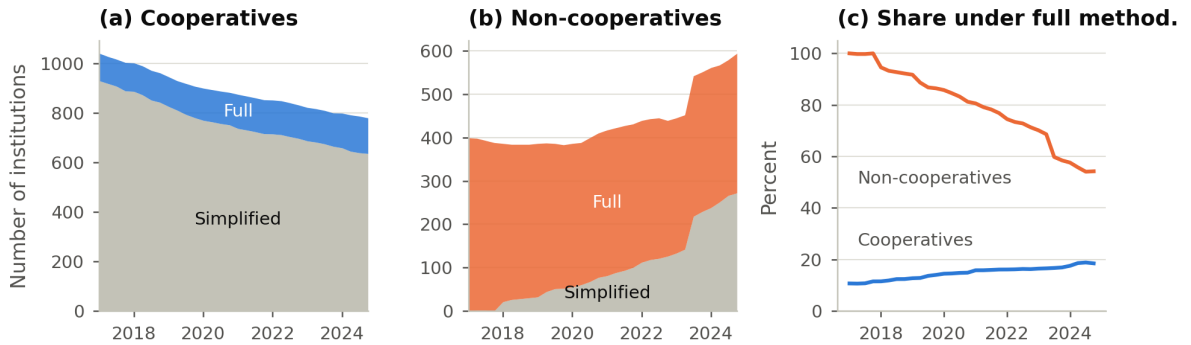


Figure 1: Institutions by prudential methodology, 2017 to 2024

Panel (c) is the share under the full methodology. The 2023 step in panel (b) is the entry of payment-institution conglomerates.

Credit cooperatives are member-owned deposit-taking institutions that lend mostly to their members. They are organised under Complementary Law 130/2009 [12]. Cooperatives held about six percent of system assets at the end of 2024, and their share has risen in every year since 2020 [13]. Much of the sector belongs to integrated systems in which central cooperatives supply liquidity facilities, consolidated risk management and a single supervisory interface to their affiliates.

1.1 Theoretical framework and expectations

Prudential capital regulation exists to internalise the externalities that a financial institution's failure imposes on depositors, on the payment system and on the safety net [1]. It also offsets the moral hazard that a deposit guarantee introduces into the institution's choice of leverage [14]. In Brazil the guarantee is mutual and split by form: bank deposits are covered by the FGC and cooperative deposits by the FGCoop, a private fund created in 2013 to which every singular cooperative must adhere, capitalised by its members at a flat rate on their deposits and able to extend them liquidity assistance [15, 16]. In the representation reading of that rationale, the supervisor acts for small, dispersed depositors who cannot monitor a leveraged intermediary; a cooperative's depositor is also a member with a residual claim and a governance right, and in the integrated systems a central body monitors on their behalf, so the problem is likely to be partly internalised. The systemic externality of failure and the moral hazard of the guarantee remain, and under a flat-rate mutual fund a thinly capitalised cooperative shifts part of its risk onto the rest of the sector. A capital template written for one form may thus bind, when applied to another, on a margin that does not carry the risk it was designed to price.

Brazil's segmentation sorts institutions by size and international activity [2], and the international surveys of proportionality record size, complexity and business model as the criteria in use, none based on organisational form [17]. Where supervisors adapt the framework to cooperatives, the adaptation concerns member shares and network arrangements [16]. The current European debate on simpler rules for smaller banks keeps to the same criteria: a German supervisory proposal reviewed by Berg and Geeroms [18] would let small, non-complex banks, the savings and cooperative banks among them, opt out of the Basel-based requirements, which the authors support only with conservative calibration and safeguards against risk-taking. Organisational form and network membership are then candidate calibration margins, and this paper supplies the profile that decides whether they point to lighter treatment or closer scrutiny.

Ownership offers a reason to expect the two forms to differ under a common methodology. A cooperative cannot raise outside voting equity as needed, so its capital is an endowment of retained earnings accumulated across member generations [19]. The member's claim carries the five property-rights problems that Cook [20] sets out: free-rider, horizon, portfolio, control and influence costs, which together weaken the incentive to build capital beyond what member service requires, since the member's claim is a claim on the use of the institution.

1.2 Comparison groups in the cooperative banking literature

Besides the proportionality work above, two strands of the literature reviewed by McKillop et al. [21] bear on this paper. The first concerns capital, profitability and stability. Hesse and Čihák [22] find in a cross-country panel that cooperative banks are more stable than commercial banks. They attribute the result to the lower volatility of cooperative returns, which more than offsets lower profitability and capitalisation. Beck et al. [23] reach a similar conclusion for German banks under one supervisory regime, where cooperative banks are farther from insolvency than both privately owned and savings banks. Ayadi et al. [24], Iannotta et al. [25] and Groeneveld [26] document the performance, risk and persistent distinctiveness of European cooperative banks, and Becchetti et al. [27] and Ferri et al. [28] show that ownership shapes lending. Cuevas

and Fischer [29] set out the tension between member orientation and prudential templates written for investor-owned banks. The second is Brazilian. Bressan et al. [30] apply the PEARLS system to Brazilian credit cooperatives and model their insolvency risk, and Carvalho et al. [31] analyse the exit and failure of Brazilian credit unions. More recently, Silva et al. [32] relate the performance of the credit unions in segment S4 to PEARLS indicators and find net lending positively associated with profitability and equity growth; they propose a comparison with the banks of the same segment, which this paper makes. Cavinato and Capitani [33] show that the social efficiency of rural credit cooperatives differs across regions. None compares cooperatives with banks under a common prudential methodology, the setting used here.

1.3 Study design and contribution

We compare cooperatives with banks across a ladder of comparison groups defined on the regulator’s own categories, sector-wide and within regional markets. The window opens in 2017Q1, the first quarter with segment reporting under Resolution 4.553, so the paper cannot speak to the crisis-period lending that Becchetti et al. [27] and Ferri et al. [28] study.

2 Aims

We ask whether organisational form leaves a measurable imprint on the balance sheet once the prudential methodology is common to both forms, and how far the answer depends on choices the analyst makes. Section 1.1 leads us to expect cooperatives to hold less regulatory capital than banks of comparable size, in a narrower band than a population that can issue equity on demand. Because the member’s claim is a claim on use, we also expect them to lend a higher share of assets. On profitability and efficiency the framework does not commit to a sign: surplus returned through pricing would depress the pre-tax return, while the cost discipline of a focused retail model would raise it. The tax exemption raises the after-tax return mechanically, which is why the two measures are reported side by side. The framework does lead us to expect lower return volatility. A cooperative that prices to members can absorb shocks through the surplus it would otherwise return, which Hesse and Čihák [22] identify as the source of cooperative stability. For provisioning and the intermediation margin it gives no reason to expect a difference. Minimum provisions are fixed by a regulatory AA to H rating grid [6], independent of ownership, so a difference on either is more likely to reflect the composition of the comparison group. These expectations follow from the framework and are stated before the results; the study was not pre-registered.

We hold no prior on the second question, which concerns method.

3 Methods

3.1 Data and classification

We construct an institution-level quarterly panel from IF.data, the public statistical repository of the Central Bank of Brazil. The summary financials, the segmentation file, the asset composition file and the income statement are merged on institution code and reporting date, and the raw

panel runs from 2014Q1 to 2024Q4. Quarter fixed effects absorb the pandemic shock, and the pre-2022 subperiod shows every surviving result (Section 4.6). All tables and figures are the authors’ calculations from the IF.data files.

The panel is IF.data’s prudential-conglomerate report, and every row carries its reporting level: 87 banks report as consolidations and 57 individually, and all cooperatives individually. Each bank takes its modal level, which is used below as a comparison-group parameter.

We classify institutions on the Central Bank’s consolidation type, *tipo de consolidado bancário* (**tcb**). The code **b3S** marks a singular credit cooperative and **b3C** a central cooperative or confederation. Consolidation type is a regulatory attribute and is stable over time; assigning each institution its modal **tcb** makes cooperative status time-invariant by construction. Central cooperatives are excluded, since they are wholesale entities that supply liquidity and risk management to affiliated singulars and carry high capital for structural reasons. By contrast, classification by name would place the sector’s own banks, Banco Cooperativo Scredi and Banco Sicoob, in the cooperative group.

3.2 Comparison groups

The cooperative side is always the singular retail cooperatives under the full methodology. The comparison side is a reported parameter, defined on the regulator’s consolidation type alone. The ladder has four rungs, and a fifth comparison keeps the bank group but matches on macro-region as well as size (Section 4.3).

- *All non-cooperatives*: every full-methodology institution outside the **b3** family.
- *Banks*, the primary group: **b1** commercial and multiple banks with a commercial portfolio and **b2** multiple and investment banks.
- *Commercial*: **b1** banks only.
- *Structural*: **b1** banks with strictly positive deposits in every quarter of their presence in the sample, defined ex ante.

Each rung can further be restricted to banks reporting individually. The ladder has no rung defined on lending, since selecting peers on an outcome would build the credit result into the comparison. Of the 339 institutions in the broad group outside the bank rung, 158 have a median of zero credit and zero funds raised (Figure 2), which makes the credit and funding differentials mechanical; we still report the broad group because its divergence from banks is one of our results.

3.3 Outcomes and two measurement corrections

We study nine quarterly outcomes, defined in Table 1, and add two stability outcomes measured at the institution level over its full-methodology quarters: the standard deviation of annualised quarterly return on assets and the z-score, the distance-to-insolvency measure used by Hesse and Čihák [22]. Both require at least eight quarters, which leaves 105 cooperatives and 137 banks, and are estimated on one row per institution with mean log assets as the size control, no quarter effects, HC3 standard errors and a wild bootstrap over institutions.

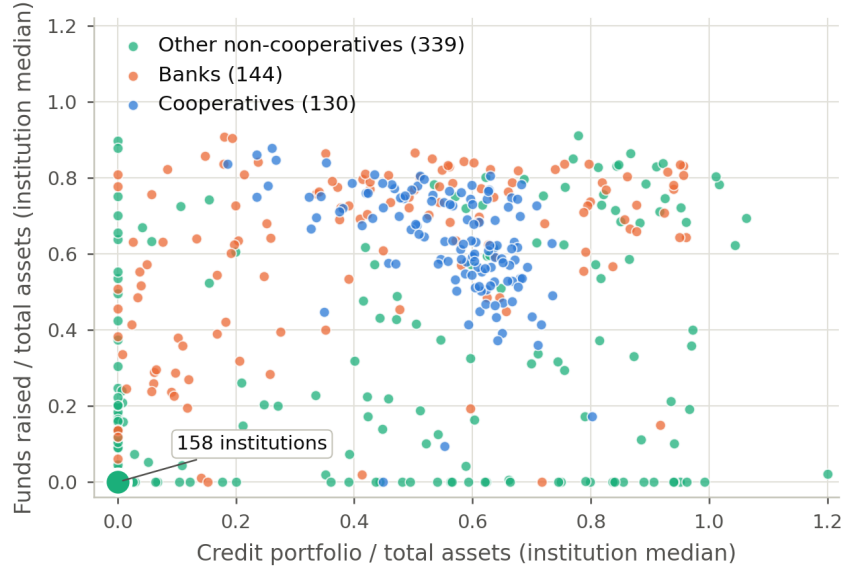


Figure 2: Credit and funding ratios of full-methodology institutions

Institution medians over 2017 to 2024. The marker at the origin stands for 158 institutions. Values above 1.2 are drawn at 1.2.

Two features of IF.data affect how the quarterly outcomes are constructed. First, income flows accumulate within the semester. Financial institutions in Brazil draw up balance sheets semiannually, in June and December [34], and the IF.data income fields reset at each date. We de-cumulate within each semester and annualise, which scales the return on assets coefficient by about 2.7 relative to reading the cumulative field as a quarterly rate, and leaves every stock-based coefficient unchanged. The de-cumulated levels agree with the Central Bank’s published figures: the median annualised return of the singular cooperatives in our panel is 2.9 percent in 2023 and 2.5 percent in 2024, against 2.9 and 2.4 in the Panorama [13]. Second, the intermediation result is reported net of loan loss provisions, because the expense subtotal (b) contains the provision line ($b5$), so $(c) = (a) + (b)$ partly reproduces the provisioning result under another name. We compute the margin gross of ($b5$), and the cost-to-income denominator likewise.

The provisioning ratio is the stock of provisions constituted under Resolution 2.682/1999 over gross credit, a weak proxy for realised delinquency, since the minimum provision follows the AA to H rating grid and the ratio reflects how the institution classifies its book.

Return on assets is measured twice. Before tax it uses line (g), the result before income tax, social contribution and profit sharing, which is the like-for-like operating measure. After tax it uses line (j), net income, which for a cooperative is the surplus before its allocation to reserves and to interest on member capital, reported separately in line (k), and is the return that accumulates as capital.

3.4 Specifications, matching and inference

For each outcome we estimate the coefficient on a cooperative dummy under four specifications: the unconditional difference in means, log total assets and quarter fixed effects, coarsened exact matching [35] with the same controls, and the same controls on the region of common support in

Table 1: Outcome definitions

Outcome	Construction
Basel capital ratio	<i>Índice de Basileia</i> as reported by BACEN, in percentage points
Leverage	Total liabilities / total assets
Return on assets, after tax	$4 \times \text{line (j)} / \text{total assets}$
Return on assets, before tax	$4 \times \text{line (g)} / \text{total assets}$
Intermediation margin	$4 \times [(c) - (b5)] / \text{total assets}$
Cost-to-income	$[(d3) + (d4)] / [(c) - (b5) + (d2)]$
Credit portfolio / assets	Credit portfolio / total assets
Provisioning ratio	Stock of provisions under Res. 2.682/1999 / gross credit
Funding ratio	<i>Captações</i> (funds raised) / total assets
Return volatility	S.d. of annualised quarterly return on assets over the institution's full-methodology quarters
Z-score	$[\text{mean return on assets} + \text{mean}(1 - \text{leverage})] / \text{s.d. of return on assets}$

IF.data lines, de-cumulated within the semester: (a) intermediation revenue, (b) intermediation expense, carried as a negative, (c) = (a) + (b) intermediation result, (b5) loan-loss provision result within (b), (d2) fee income, (d3) personnel expense, (d4) administrative expense, (g) result before taxes and profit sharing, (j) net income.

log assets. Cooperative status is fixed within an institution and collinear with institution fixed effects, so the estimates are conditional correlations.

The coarsened-exact-matching estimand weights control units so that the treated-to-control ratio is equal within every stratum [35]; an unweighted regression on the matched sample recovers a different estimand, so we report both. We coarsen on size quintile and control for region. Region-exact matching would leave too few controls, since cooperatives concentrate in the South and banks in the Southeast: once unequal weights are discounted, the effective control sample is about eighteen institutions, against seventy-six under size coarsening. Region-exact estimates are reported as a second parameter (Section 4.3).

Standard errors are clustered by institution, since cooperative status is assigned at the institution level and observations are serially correlated [36]. Clustered inference is unreliable when few clusters are treated [37], as in the system-by-system estimates, so every coefficient also carries a restricted wild cluster bootstrap, which re-draws the sign of each institution's residuals under the null and compares the observed statistic with that distribution [38]. In simulations with six treated clusters, the size of the smallest system split (online appendix), the clustered test over-rejects and the bootstrap is conservative at one percent; the problem concerns the system splits, while the main comparison rests on about 130 treated clusters, where the two agree. Quantile coefficients carry intervals from an institution bootstrap. Several outcomes are right-skewed, so we report the conditional median alongside the conditional mean, under one rule for every outcome: tiers are assigned on the conditional means alone, and when the quantile profile shows one tail driving the mean, as for capital, cost-to-income and the pre-tax return, the magnitude is read at the median while the tier records the mean result.

We classify each outcome by an explicit rule, reported in the last column of Table 2 and stated in full in the online appendix. An outcome is *null* unless it is significant at five percent, with one sign, in all four specifications; it is *stable* if it also survives every winsorisation rule within a factor of two and its average quantile coefficient between the twentieth and eightieth

percentiles is at least half the conditional mean, and *stable in sign* otherwise.

4 Results

The analysis sample contains 130 singular cooperatives and 144 banks, 7,234 institution-quarters, from 2017Q1 to 2024Q4. Ninety percent of cooperative observations fall inside the common support region of log assets (Figure 3), against forty-eight percent of bank observations. Matching and trimming bring the standardised difference in log assets from -0.72 to close to zero. Unconditional means and medians are in the online appendix.¹

Three statements summarise what follows. First, under the same prudential rules the largest credit cooperatives run at lower cost and post steadier returns than banks of the same size, lend a larger share of their assets than banks nationally, and keep about two percentage points more of their return after tax, of which the tax exemption explains about two fifths; what they keep is the only capital they can build. Second, their capital ratios sit in a narrow band: below the bank average, which a few small and heavily capitalised banks pull up, and above banks at the same quantile at the thin end of the distribution, where supervisors look. Third, two often-cited differences, lower provisioning and thinner margins, vanish once cooperatives are compared with banks alone.

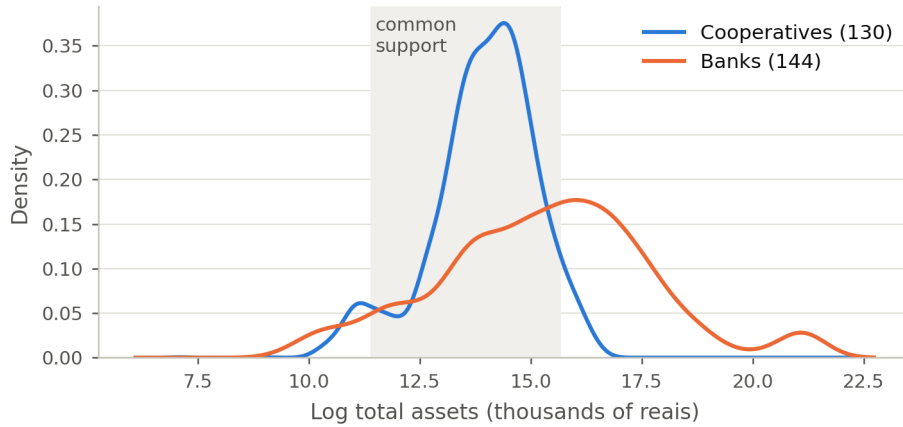


Figure 3: Log total assets, cooperatives and banks

4.1 The cooperative differential

Table 2 reports the coefficient on the cooperative dummy under the four specifications, together with the conditional median counterpart of the second column and the retention ratio ρ , the share of the raw differential that survives the restriction to common support.

Five of the nine quarterly outcomes keep their sign and significance across all four specifications and every winsorisation rule, including no winsorisation, and so do both stability outcomes. By the rule of Section 3.4, credit intensity and the after-tax return are stable among the quarterly outcomes, capital, leverage and cost-to-income are stable in sign, and the pre-tax

¹The online appendix (supplement.pdf) is at <https://github.com/r2nery/coops-basel-regulation>.

Table 2: Cooperative differential against banks

Outcome	(1) Raw	(2) Controls	(3) CEM	(4) Trim	Median (2)	ρ	Tier
Basel capital ratio (pp)	-8.19***	-15.77***	-17.50***	-14.21***	-2.18	1.73	stable in sign
Leverage	0.0561**	0.1220***	0.1046***	0.0813***	0.0283	1.45	stable in sign
Return on assets, after tax	0.0181***	0.0215***	0.0179***	0.0159***	0.0171	0.88	stable
Return on assets, before tax	0.0118***	0.0132**	0.0088	0.0062	0.0144		null
Cost-to-income ratio	-0.6025***	-0.7923***	-0.8385***	-0.7522***	-0.1796	1.25	stable in sign
Credit portfolio / assets	0.1654***	0.1881***	0.1701***	0.1780***	0.2448	1.08	stable
Provisioning ratio	0.0020	-0.0012	0.0010	0.0009	0.0236		null
Intermediation margin	-0.0003	-0.0190	-0.0107	-0.0171	0.0203		null
Funding ratio	0.0461	0.0887**	0.0536	0.0330	-0.0247		null
Return on assets volatility	-0.0110***	-0.0189***	-0.0175***	-0.0170***	-0.0133	1.55	stable
Z-score	4.20***	6.32***	6.14***	6.20***	7.58	1.48	stable

Standard errors clustered by institution, HC3 for the last two rows; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, confirmed by a wild cluster bootstrap. Median (2): conditional median of column (2). $\rho = |b_{\text{trim}}|/|b_{\text{raw}}|$. Tier: see Section 3.4.

return, provisioning, the margin and funding are null; return volatility and the z-score are both stable.

Cooperatives allocate a larger share of the balance sheet to credit, by nearly a fifth of total assets under controls and by more at the conditional median.

Profitability splits into the two channels of Section 1. Before tax, the cooperative differential is positive in every specification but significant only in the first two, so by the rule it is null. The null comes from the mean: the conditional median differential before tax is 1.4 percentage points with a bootstrap interval that excludes zero, and the quantile coefficient turns negative only in the top decile (Table 3), where a high-earning bank tail pulls the mean towards zero. The typical cooperative thus earns more than a comparable bank before tax, while the mean gap is not established. After tax the differential is about two percentage points under controls, significant everywhere and stable. The distance between the two rows is the tax exemption: among quarters with a positive pre-tax result, the median effective rate of income tax and social contribution is about a third for banks and close to zero for cooperatives, and the pre-tax differential under controls is about three fifths of the after-tax one, so tax accounts for about two fifths of the after-tax gap. The after-tax row is what a cooperative retains and accumulates as capital; the pre-tax row is the operating comparison.

Cooperatives operate at lower cost, although the size of the difference is fragile. The controlled coefficient roughly triples between tight winsorisation and none, because the denominator approaches zero for banks that barely intermediate. We rely on the conditional median, whose bootstrap interval excludes zero.

The capital result requires the most care. Unconditionally, the median cooperative is better capitalised than the median bank for most of the window (Figure 4). Yet the size-controlled mean differential is large and negative in every column of Table 2, and under controls about seven times the conditional median. Figure 5 reconciles the two: the cooperative coefficient is positive at the bottom of the conditional capital distribution and negative at the top, and it crosses zero near the fortieth percentile. Averaged over the twentieth to eightieth percentiles, the differential is about a third of the conditional mean (Table 3). Unconditionally, 14 percent of bank-quarters exceed a Basel ratio of 50 percent, against 1 percent of cooperative-quarters. At

equal size, what mainly separates the two groups is dispersion: residualised on size and quarter, the standard deviation of the capital ratio among cooperatives is about a third of the bank figure (online appendix; Figure 4).

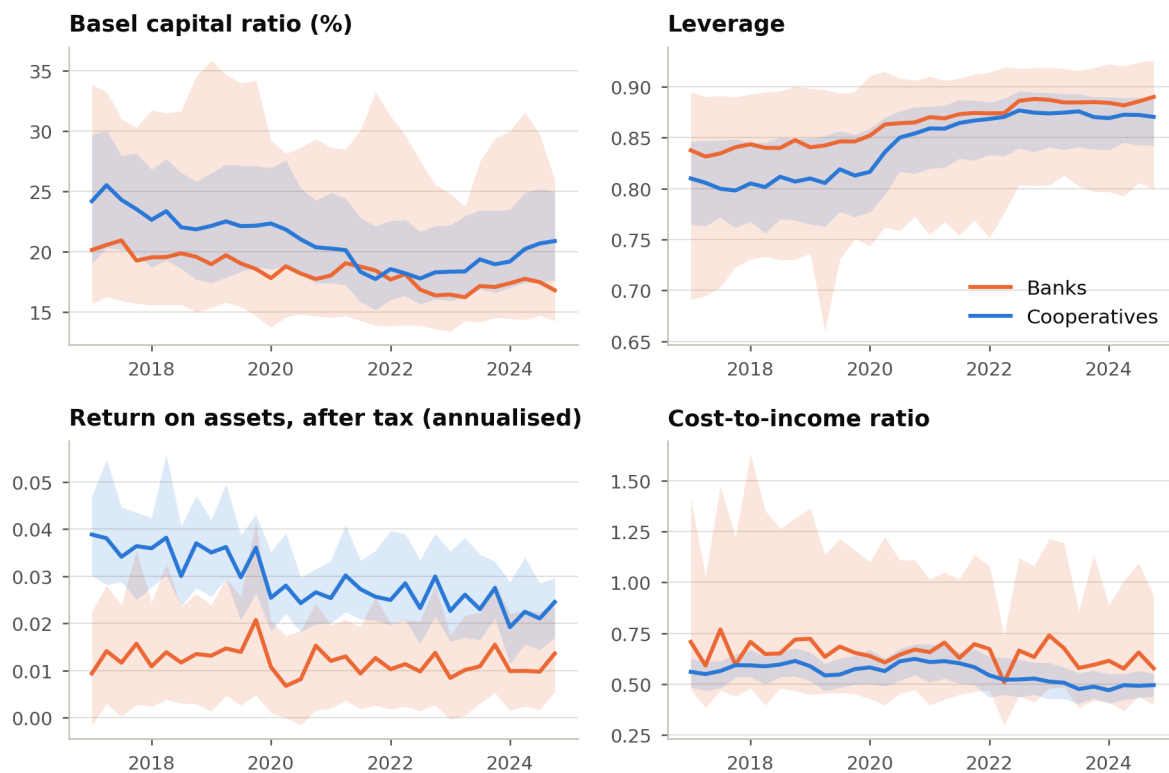


Figure 4: Medians and interquartile ranges, 2017 to 2024

Two features locate the bank upper tail. It sits among the banks that report individually, against which the controlled gap is about four times the gap against the consolidations (online appendix), and it thins when banks must be deposit-funded in every quarter: the structural rung roughly halves both the share of bank-quarters above 50 percent and the estimated mean gap, which remains significant (Figure 6). Against deposit-taking banks the capital difference is thus a modest gap in level and a large gap in dispersion.

At equal size, cooperatives are the more homogeneous group on every outcome, with residual standard deviations between a tenth and a half of the bank figure. Leverage and capital are closely correlated among cooperatives, and we treat the two as a single result.

Cooperative returns are also less volatile. At comparable size cooperatives show a lower standard deviation of return on assets and a higher z-score, in all four designs and on every national bank rung. The two measures share the standard deviation of return on assets, so we treat them as one result. Taxes compress the after-tax volatility of bank returns, and before tax both measures keep their sign and significance, with the median volatility gap nearly doubling (Table A11 of the online appendix).

Table 3: Quantile regression coefficients

Outcome	q_{10}	q_{25}	q_{50}	q_{75}	q_{90}	Mean
Basel ratio (pp)	2.81 [2.06, 3.75]	1.61 [0.19, 2.76]	-2.18 [-6.52, 1.43]	-14.23 [-22.45, -8.20]	-34.37 [-47.28, -21.82]	-15.77 [-21.93, -9.62]
Leverage	0.294 [0.194, 0.370]	0.117 [0.069, 0.174]	0.028 [0.003, 0.062]	-0.009 [-0.021, 0.008]	-0.023 [-0.037, -0.007]	0.122 [0.082, 0.162]
Return on assets, after tax	0.044 [0.032, 0.062]	0.026 [0.020, 0.030]	0.017 [0.014, 0.021]	0.010 [0.005, 0.015]	-0.006 [-0.017, 0.004]	0.021 [0.014, 0.029]
Return on assets, before tax	0.052 [0.038, 0.066]	0.028 [0.023, 0.035]	0.014 [0.009, 0.019]	-0.003 [-0.011, 0.005]	-0.029 [-0.046, -0.016]	0.013 [0.004, 0.023]
Cost-to-income	0.110 [0.009, 0.151]	0.006 [-0.051, 0.067]	-0.180 [-0.277, -0.107]	-0.647 [-0.870, -0.438]	-1.756 [-2.822, -1.158]	-0.792 [-1.047, -0.538]
Credit / assets	0.383 [0.343, 0.428]	0.424 [0.370, 0.466]	0.245 [0.146, 0.363]	-0.019 [-0.085, 0.069]	-0.170 [-0.219, -0.111]	0.188 [0.129, 0.248]
Provisioning ratio	0.030 [0.026, 0.033]	0.030 [0.026, 0.034]	0.024 [0.018, 0.029]	0.001 [-0.016, 0.015]	-0.054 [-0.093, -0.017]	-0.001 [-0.015, 0.013]

Quantile regressions on the cooperative dummy, log assets and quarter effects; the last column is OLS. Brackets: 95 percent institution-bootstrap intervals (199 draws) and the clustered interval for the mean.

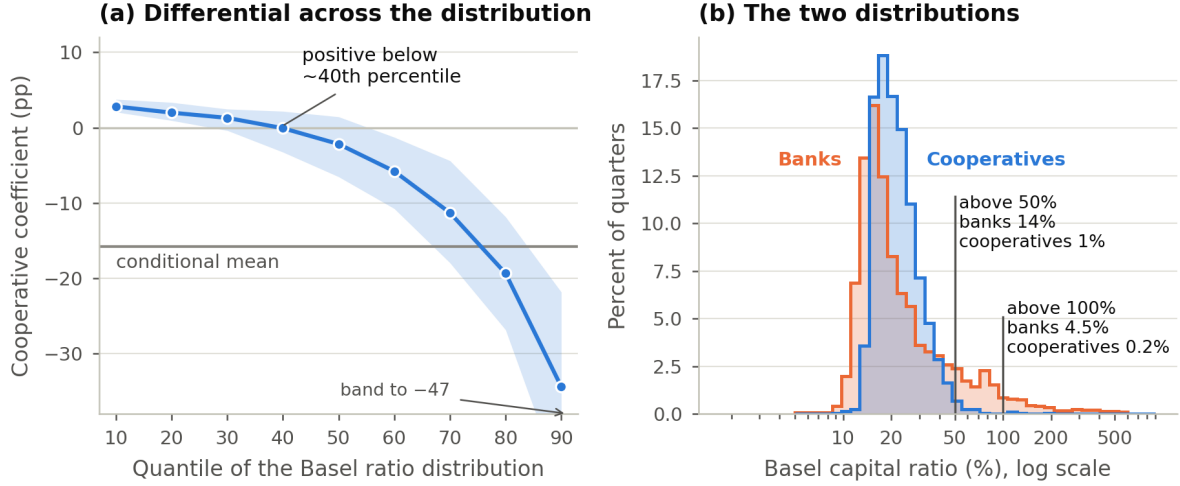


Figure 5: The capital ratio across the distribution

Panel (a): quantile regression coefficients with a 95 percent institution-bootstrap band, and the conditional mean. Panel (b): Basel ratio before winsorisation, on a log scale; values below 5 percent are drawn at 5, and bar heights are percent of each group's quarters.

4.2 Comparison-group dependence

Table 4 states which differentials survive against which comparison group, and Figure 6 shows their size and precision on the same five rungs: the broad group, banks, banks within macro-region, commercial banks and structural banks. The after-tax return, cost-to-income and return volatility survive on every rung; credit intensity survives on the four national rungs and not within region (Section 4.3); the pre-tax return survives on the three bank rungs only. Two results that appear against the broad group disappear against banks.

Table 4: What survives against which comparison group

Outcome	All non-coop.	Banks	Within region	Commercial	Structural
Basel capital ratio (pp)	—	—	—	—	—
Leverage	+	+	+	+	+
Return on assets, after tax	+	+	+	+	+
Return on assets, before tax	·	+	·	+	+
Cost-to-income ratio	—	—	—	—	—
Credit portfolio / assets	+	+	·	+	+
Provisioning ratio	—	·	·	·	·
Intermediation margin	—	·	·	·	+
Funding ratio	+	+	·	·	·
Return on assets volatility	—	—	—	—	—
Z-score	+	+	·	+	+

Sign of the cooperative coefficient where it is significant at five percent, a dot otherwise. Controlled specification; within region uses the matched specification. HC3 errors for the last two rows.

Against all non-cooperatives, cooperatives appear to provision less, significant at one percent in all four specifications, while against banks the coefficient is close to zero. The broad-group gap comes from the non-banking lenders, whose provisioning ratios are nearly three times those of banks and cooperatives. At the conditional median the sign reverses, with cooperatives provisioning more. Split into terciles of the Basel ratio, no tercile shows cooperatives provisioning less.

The intermediation margin behaves in the same way: large and highly significant against the broad group, small and insignificant against banks, positive and significant against structural banks, and positive at the conditional median, so its sign depends on the comparison group and on the statistic and we keep it as a null. The funding ratio is significant in one specification only and is left out of the findings.

4.3 Credit intensity within and across regional markets

Matched on size alone, cooperatives allocate a substantially larger share of assets to credit than comparable banks, significant at one percent in every specification. Matched also on macro-region, the differential is small and indistinguishable from zero under any coarsening scheme. Which estimate is the parameter depends on an untestable assumption: if cooperatives concentrate where they do for reasons internal to member-based finance, region is a mediator and the size-only estimate is the parameter; if regional market structure is unrelated to form, region is a confounder and credit intensity is a null. We report both and label every claim by

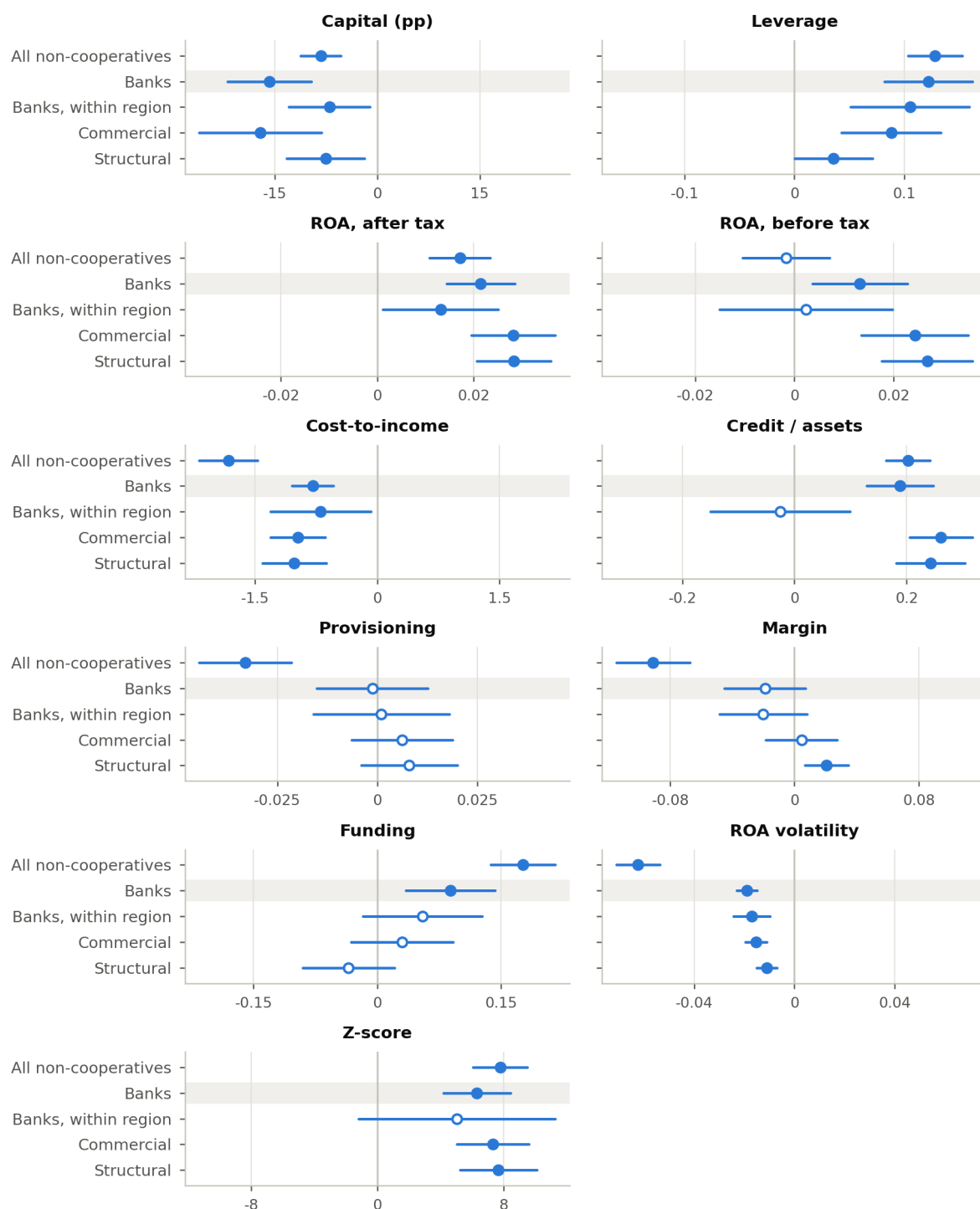


Figure 6: Cooperative coefficient by comparison group

Controlled specification; the within-region rung also matches on macro-region. Intervals are 95 percent, clustered by institution (HC3 for the stability outcomes). Filled markers: $p < 0.05$. Shaded: banks, the primary group.

the comparison it rests on.

With an effective control sample of eighteen, the within-market null may reflect low power as much as the parameter itself. Return volatility survives the within-region comparison, and the z-score does not; we read the pair as one result whose within-market status is uncertain.

4.4 Network-level heterogeneity

The online appendix splits the cooperative side by system. Sicredi and Sicoob account for all but seven of the cooperatives, one of which belongs to Cresol, so the split describes those two networks; a test of network affiliation as such is beyond it. Cooperatives in both networks hold less regulatory capital than comparable banks, significant under the bootstrap in each. The six independents contribute five treated clusters, since one lacks the outcome, and their capital estimate is positive and indistinguishable from noise.

The split locates the pooled capital differential in the two large networks without explaining it. A network backstop is one candidate, since centralised liquidity and consolidated risk management may let an affiliate run a thinner individual cushion, but the networks also differ in the size, geography, business mix and governance of their affiliates, and the split leaves the backstop entangled with these. Provisioning is insignificant in every system.

The comparison also crosses a boundary: banks enter partly as prudential conglomerates, cooperatives as singulars, and the centrals are excluded. IF.data ties a singular to its system but not to its central, so the asymmetry can only be bounded: at the end of 2024 the centrals held about a quarter of the singulars' assets, mostly the singulars' own centralised deposits, but about six percent of their regulatory capital and a tenth of their administrative and personnel expenses, so consolidation would likely move the capital and cost differentials by far less than their size.

4.5 Selection into the analysis sample

The full-methodology restriction equalises the regulatory environment, but it also selects the sample. The cooperatives we observe are the largest in the country, about a tenth of all cooperatives recorded in the panel. Fifty-one cooperatives are observed under the simplified methodology before adopting the full one; we compare them with cooperatives that never adopt, on pre-adoption quarters only, conditioning on size and quarter. The two groups show no difference in leverage, but adopters were less credit-intensive (-0.068 , $p = 0.002$), marginally less profitable and marginally leaner. This bears on generalisation rather than on the estimates: they describe the adopters, and the wider cooperative population would, if anything, show larger credit and profitability differentials and a smaller efficiency one.

4.6 Sensitivity checks

The online appendix reports the conditional medians and the specification without the eleven S1 and S2 banks, which moves no significant coefficient by more than two percent. It also reports a split at the January 2022 replacement of the 2013 capital rules. Every surviving result keeps its sign and significance in both subperiods, and provisioning and the margin remain null in

both. Magnitudes are roughly a fifth smaller after 2022. Four winsorisation rules, dropping the sector’s two cooperative-owned banks and dropping the six cooperatives that revert to the simplified methodology leave every surviving sign unchanged.

5 Discussion

The surviving results describe a focused retail lender, consistent with a member-service orientation in which capital accumulation is a means to member credit [19]. For supervisors, the findings suggest a benchmark calibrated on lending institutions and a reading of the whole distribution as well as the mean.

The capital results also qualify the property-rights argument. Cook [20] predicts weak incentives to build capital, yet the median cooperative is better capitalised than the median bank and holds more capital at the thin end of the distribution. The property-rights problems bear on capital raised from members, which a cooperative cannot expand at will; the reserves it retains are indivisible and accumulate across member generations [19]. What the argument rules out is the upper tail, since a cooperative has no reason to hold capital far above what its lending requires and no outside investor to supply it. The narrow band appears to be the profile that endowment capital predicts, and the mean gap the absence of that tail.

Our results agree with Hesse and Čihák [22] on two of the three bank-level differences they report, lower capitalisation at comparable size and lower return volatility. They differ on the third. In their sample cooperative stability came with lower profitability, whereas in this population the typical cooperative earns more than a comparable bank before tax, retains more after it, and is less volatile on both measures. Part of the after-tax difference is the tax exemption and the pre-tax mean gap is not established, so the disagreement is narrower than the after-tax figures suggest; the rest may reflect our population of the largest cooperatives.

For supervision, the relevant combination is capital and provisioning. At the thin end of the capital distribution, where prudential concern attaches, cooperatives hold more capital than comparable banks and provision no less. Several differentials that look large at the mean are differentials in dispersion, so we weight the conditional median and the quantile profile as much as the conditional mean.

Three analyst choices determined whether headline results existed: the comparison group, which as “all non-cooperatives” imports some 160 institutions that neither lend nor raise funds; the reading of IF.data income fields as quarterly flows; and unweighted OLS on the matched sample. Some disagreements in the literature may reflect such choices.

5.1 Limitations

Because cooperative status is time-invariant, the estimates are conditional correlations. The full-methodology restriction selects the largest cooperatives, and the results generalise only to that subpopulation. The provisioning ratio is an accounting variable and does not measure delinquency. The stability outcomes are institution-level statistics on a subsample, estimated without quarter effects; the panel is unbalanced, an institution that is incorporated or closes leaves it at its last reported quarter, and institutions with fewer than eight quarters are absent

from that subsample. Most systems are too thinly represented for separate inference, and the network backstop hypothesis needs consolidated network data that IF.data does not carry. The mediator question in Section 4.3 also needs a market measure finer than macro-region, and IF.data’s credit-modality modules, which carry rural credit, are the natural next step, since the social efficiency of rural cooperatives differs across regions [33]. Finally, a within-institution design around adoption of the full methodology fails to identify a regime effect, because adoption follows growth and pre-adoption trends run through the adoption date without a break.

6 Contributions and Implications

This paper characterises cooperative behaviour under full Basel-style regulation in a large emerging market, on the full population of cooperatives and banks under the full methodology. The substantive contribution is this profile, together with two nulls that correct claims the same data appear to support under a looser comparison group. The methodological contribution is a protocol that defines peers on the regulator’s own categories, reports a ladder of comparison groups, and separates sector-wide from within-market differences. It also states how cumulative flows were converted to rates and reports the matched estimand; each step changed at least one conclusion.

Our results bear on two margins of proportionality beyond size [2], both framed as questions since the evidence is descriptive. The first is organisational architecture, since the capital differential is a property of the two large networks. If the backstop hypothesis is right, a proportional framework might treat an integrated network and its backstop as the unit for parts of the capital and liquidity assessment, as Coelho et al. [16] report for the French cooperative groups. If the differential reflects the affiliates’ size, geography or business mix, that suggestion loses its basis. The second is the calibration of the benchmark itself. A benchmark set against the average full-methodology institution may describe the cooperative configuration poorly, because that average comes from a distribution whose upper region cooperatives do not inhabit. Section 1.1 indicates why a lighter requirement does not follow automatically. The representation rationale weakens for a member-owned, network-monitored cooperative, but the systemic externality and the guarantee remain, and the FGCoop’s flat-rate mutual funding places the cost of a thin cushion on the rest of the sector, so thinner capital may call for closer supervisory attention as well as for relief.

The fiscal treatment of cooperative acts is the rule difference the common methodology leaves in place, and the results locate its effect. It appears in the retained return, the only capital a cooperative can build, and accounts for about two fifths of the after-tax differential; the rest appears before tax, at the median though not at the mean, on an intermediation margin that does not differ at the mean. At the banks’ median effective rate the cooperatives’ median after-tax return would fall from about 2.9 to about 2 percent, and their capital would accumulate about three tenths more slowly, assuming an unchanged payout of surplus to members. Whether the exemption is a subsidy, since it is forgone public revenue, or the recognition of mutual self-help, since the sector also funds its own guarantee through the FGCoop, is a question these data do not settle.

For the cooperative movement, the difference between the two forms persists under a common prudential methodology, in a narrower form than a loose comparison suggests. Our evidence does not support a case resting on cooperatives provisioning more conservatively than banks. A case resting on cost at the median and on return stability holds against every definition of a bank we tried; one resting on profitability holds for the retained return and, for the operating return, at the median rather than at the mean. One resting on credit intensity holds nationally and remains unshown within regional markets; the territorial claim is the one these data leave open.

Declaration

The authors work at the Organization of Brazilian Cooperatives (OCB System), which represents the cooperative sector in Brazil. The study uses public data and a public reproduction pipeline, and the views expressed are the authors' own.

7 Key References

- [1] Basel Committee on Banking Supervision. Basel III: A global regulatory framework for more resilient banks and banking systems. Technical report, Bank for International Settlements, Basel, 2011. December 2010, revised June 2011.
- [2] Conselho Monetário Nacional. Resolução nº 4.553, de 30 de janeiro de 2017. Segmentação do conjunto das instituições financeiras para fins de aplicação proporcional da regulação prudencial. Banco Central do Brasil, 2017. URL https://normativos.bcb.gov.br/Lists/Normativos/Attachments/50335/Res_4553_v2_L.pdf.
- [3] Conselho Monetário Nacional. Resolução nº 4.606, de 19 de outubro de 2017. Metodologia facultativa simplificada para apuração do requerimento mínimo de Patrimônio de Referência Simplificado (PRS5). Banco Central do Brasil, 2017. URL https://normativos.bcb.gov.br/Lists/Normativos/Attachments/50454/Res_4606_v5_P.pdf.
- [4] Conselho Monetário Nacional. Resolução nº 4.955, de 21 de outubro de 2021. Metodologia para apuração do Patrimônio de Referência. Banco Central do Brasil, 2021.
- [5] Conselho Monetário Nacional. Resolução nº 4.958, de 21 de outubro de 2021. Requerimentos mínimos de Patrimônio de Referência, de Nível I e de Capital Principal e adicional de Capital Principal. Banco Central do Brasil, 2021.
- [6] Conselho Monetário Nacional. Resolução nº 2.682, de 21 de dezembro de 1999. Critérios de classificação das operações de crédito e regras para constituição de provisão. Banco Central do Brasil, 1999. URL https://www.bcb.gov.br/pre/normativos/res/1999/pdf/res_2682_v2_L.pdf.
- [7] Conselho Monetário Nacional. Resolução nº 4.401, de 27 de fevereiro de 2015. Requerimentos mínimos de liquidez de curto prazo (LCR). Banco Central do Brasil, 2015.

- [8] Conselho Monetário Nacional. Resolução nº 4.616, de 30 de novembro de 2017. Requerimento mínimo de liquidez de longo prazo (NSFR). Banco Central do Brasil, 2017. URL https://normativos.bcb.gov.br/Lists/Normativos/Attachments/50474/Res_4616_v1_0.pdf.
- [9] Conselho Monetário Nacional. Resolução nº 4.615, de 30 de novembro de 2017. Razão de Alavancagem. Banco Central do Brasil, 2017.
- [10] Brasil. Lei nº 5.764, de 16 de dezembro de 1971. Define a Política Nacional de Cooperativismo, institui o regime jurídico das sociedades cooperativas. Diário Oficial da União, 1971. URL https://www.planalto.gov.br/ccivil_03/leis/15764.htm.
- [11] Brasil. Lei nº 10.865, de 30 de abril de 2004. Dispõe sobre a Contribuição para os Programas de Integração Social e de Formação do Patrimônio do Servidor Público e a Contribuição para o Financiamento da Seguridade Social incidentes sobre a importação de bens e serviços. Diário Oficial da União, 2004. URL https://www.planalto.gov.br/ccivil_03/_ato2004-2006/2004/lei/110.865.htm.
- [12] Brasil. Lei Complementar nº 130, de 17 de abril de 2009. Dispõe sobre o Sistema Nacional de Crédito Cooperativo. Diário Oficial da União, 2009. URL https://www.planalto.gov.br/ccivil_03/leis/lcp/lcp130.htm.
- [13] Banco Central do Brasil. Panorama do Sistema Nacional de Crédito Cooperativo: data-base dezembro de 2024. Banco Central do Brasil, June 2025.
- [14] Douglas W. Diamond and Philip H. Dybvig. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419, 1983. doi: 10.1086/261155.
- [15] Conselho Monetário Nacional. Resolução nº 4.284, de 5 de novembro de 2013. Aprova o Estatuto e o Regulamento do Fundo Garantidor do Cooperativismo de Crédito (FGCoop) e estabelece a forma de contribuição. Banco Central do Brasil, 2013. URL https://www.bcb.gov.br/pre/normativos/res/2013/pdf/res_4284_v1_o.pdf.
- [16] Rodrigo Coelho, José Alberto Mazzillo, Jean-Philippe Svoronos, and Tommy Yu. Regulation and supervision of financial cooperatives. FSI Insights on Policy Implementation 15, Bank for International Settlements, Financial Stability Institute, Basel, 2019.
- [17] Basel Committee on Banking Supervision. Proportionality in bank regulation and supervision: A survey on current practices. Technical report, Bank for International Settlements, Basel, 2019.
- [18] Jesper Berg and Hans Geeroms. Calls for EU financial-rule simplification should not spill over into lower standards. Bruegel Analysis 21/2026, July 2026. URL <https://www.bruegel.org/analysis/calls-eu-financial-rule-simplification-should-not-spill-over-lower-standards>.
- [19] Wim Fonteyne. Cooperative banks in Europe: Policy issues. IMF Working Paper WP/07/159, International Monetary Fund, Washington, DC, 2007.

- [20] Michael L. Cook. The future of U.S. agricultural cooperatives: A neo-institutional approach. *American Journal of Agricultural Economics*, 77(5):1153–1159, 1995. doi: 10.2307/1243338.
- [21] Donal McKillop, Declan French, Barry Quinn, Anna L. Sobiech, and John O. S. Wilson. Cooperative financial institutions: A review of the literature. *International Review of Financial Analysis*, 71:101520, 2020. doi: 10.1016/j.irfa.2020.101520.
- [22] Heiko Hesse and Martin Čihák. Cooperative banks and financial stability. IMF Working Paper WP/07/2, International Monetary Fund, Washington, DC, 2007.
- [23] Thorsten Beck, Heiko Hesse, Thomas Kick, and Natalja von Westernhagen. Bank ownership and stability: Evidence from Germany. Unpublished working paper, Deutsche Bundesbank, Tilburg University and International Monetary Fund, April 2009, 2009.
- [24] Rym Ayadi, David T. Llewellyn, Reinhard H. Schmidt, Emrah Arbak, and Willem Pieter De Groen. Investigating diversity in the banking sector in Europe: Key developments, performance and role of cooperative banks. Technical report, Centre for European Policy Studies, Brussels, 2010.
- [25] Giuliano Iannotta, Giacomo Nocera, and Andrea Sironi. Ownership structure, risk and performance in the European banking industry. *Journal of Banking & Finance*, 31(7): 2127–2149, 2007. doi: 10.1016/j.jbankfin.2006.07.013.
- [26] Hans Groeneveld. Reconciling different truths about isomorphic pressure and distinctive behavior at European cooperative banks: Back to the future with Raiffeisen’s principles. *Annals of Public and Cooperative Economics*, 91(3):359–386, 2020. doi: 10.1111/apce.12280.
- [27] Leonardo Becchetti, Rocco Ciciretti, and Adriana Paolantonio. The cooperative bank difference before and after the global financial crisis. *Journal of International Money and Finance*, 69:224–246, 2016. doi: 10.1016/j.jimonfin.2016.06.016.
- [28] Giovanni Ferri, Panu Kalmi, and Eeva Kerola. Does bank ownership affect lending behavior? Evidence from the Euro area. *Journal of Banking & Finance*, 48:194–209, 2014. doi: 10.1016/j.jbankfin.2014.05.007.
- [29] Carlos E. Cuevas and Klaus P. Fischer. Cooperative financial institutions: Issues in governance, regulation, and supervision. World Bank Working Paper 82, The World Bank, Washington, DC, 2006.
- [30] Valéria Gama Fully Bressan, Marcelo José Braga, Aureliano Angel Bressan, and Moisés de Andrade Resende Filho. Uma aplicação do sistema PEARLS às cooperativas de crédito brasileiras. *Revista de Administração (RAUSP)*, 46(3):258–274, 2011. doi: 10.5700/rausp1011.
- [31] Flávio Leonel de Carvalho, Maria Dolores Montoya Diaz, Sigismundo Bialoskorski Neto, and Aquiles Elie Guimarães Kalatzis. Exit and failure of credit unions in Brazil: A risk analysis. *Revista Contabilidade & Finanças*, 26(67):70–84, 2015. doi: 10.1590/1808-057x201411390.

- [32] Alexsandra Silva, Jos ete Florencio dos Santos, and Adhemar Ranciaro Neto. Performance of Brazilian credit unions: An analysis from PEARLS indicators. *RAM. Revista de Administra  o Mackenzie*, 24(1):eRAMR230057, 2023. URL <https://www.scielo.br/j/ram/a/gv8jtmTShyVDhNzwcKRgDzj/>.
- [33] Nayara Regina Cavinato and Daniel Henrique Dario Capitani. Efici ncia social das cooperativas de cr dito rural no Brasil: uma an lise sob a  tica regional. *Economia e Sociedade*, 32(2):429–455, 2023. doi: 10.1590/1982-3533.2023v32n2art08.
- [34] Brasil. Lei n  4.595, de 31 de dezembro de 1964. Disp e sobre a pol tica e as institui  es monet rias, banc rias e credit cias, cria o Conselho Monet rio Nacional e d  outras provid ncias. Di rio Oficial da Uni o, 1964. URL https://www.planalto.gov.br/ccivil_03/leis/14595.htm.
- [35] Stefano M. Iacus, Gary King, and Giuseppe Porro. Causal inference without balance checking: Coarsened exact matching. *Political Analysis*, 20(1):1–24, 2012. doi: 10.1093/pan/mpr013.
- [36] A. Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.
- [37] James G. MacKinnon and Matthew D. Webb. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2017. doi: 10.1002/jae.2508.
- [38] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.